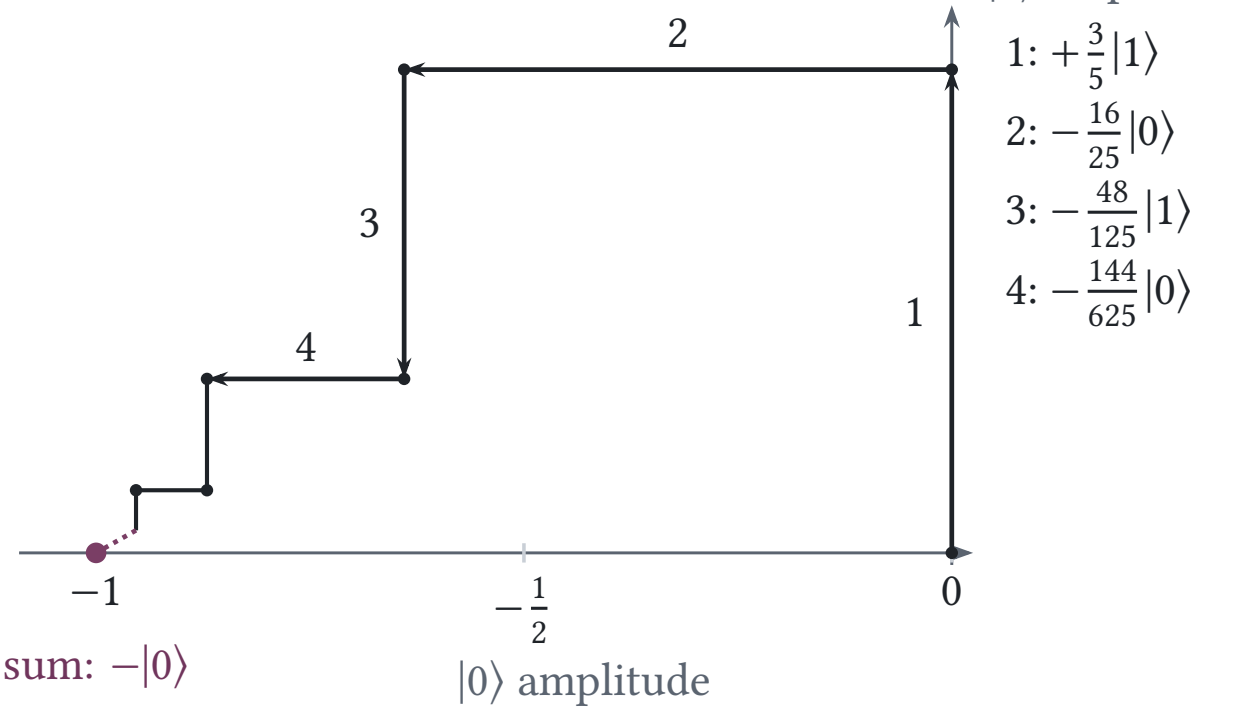
*(a)* `trace`*: add the exit vectors*




$|1\rangle$ amplitude

1: $+\frac{3}{5}|1\rangle$

2: $-\frac{16}{25}|0\rangle$

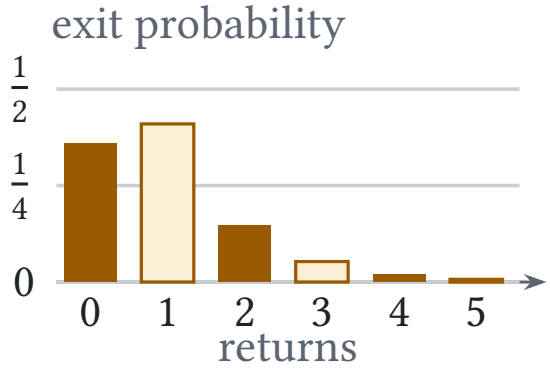
3: $-\frac{48}{125}|1\rangle$

4: $-\frac{144}{625}|0\rangle$

sum: $-|0\rangle$

each arrow adds one exit
intermediate points are unnormalized sums

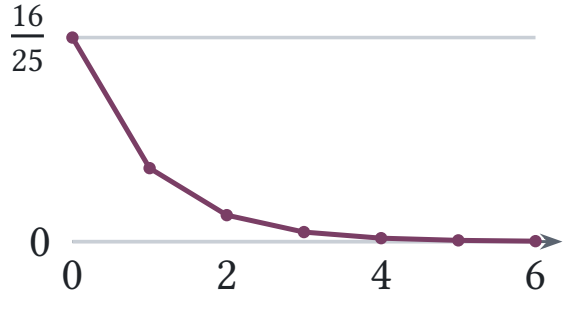
*(b)* `while`*: add exit probabilities*




even total: 9/17 on $|1\rangle$

odd total: 8/17 on $|0\rangle$

*(c) probability still inside*




after $k+1$ rounds
each further round retains 9/25